

Retail Business Performance & Profitability Analysis

Introduction

Retail businesses generate large amounts of transactional data that can be analyzed to understand sales performance, profitability, and product behavior. This project focuses on analyzing retail sales data to identify profitable categories, loss-making products, and areas where business decisions can be improved.

Abstract

The objective of this project is to analyze retail transaction data and uncover insights related to sales, profit, product performance, and category-wise profitability. The analysis helps identify profit-draining products, high-performing categories, and opportunities for improving business strategies.

The project uses data analysis techniques and visualization to transform raw sales data into meaningful business insights. The dashboard provides an interactive view of key performance indicators and product-level analysis.

Tools Used

- **Python (Pandas)** – Data cleaning and exploratory data analysis
- **SQL (MySQL Workbench)** – Data querying and business calculations
- **Power BI** – Dashboard creation and visualization
- **CSV Dataset** – Retail transaction data source

Steps Involved in Building the Project

1. Data Preparation

- Imported retail transaction data.
- Checked dataset structure and columns.
- Cleaned and prepared data for analysis.

2. Data Analysis Using Python

- Analyzed sales and profit patterns.
- Calculated profit margins.
- Studied category-wise and product-wise performance.
- Identified relationships between sales factors and profitability.

3. SQL Analysis

Created SQL queries to:

- Check missing records.
- Calculate total sales and profit.
- Analyze profit margin by category.
- Compare profit across sub-categories.
- Identify top-performing and loss-making products.

4. Dashboard Development

Built an interactive dashboard containing:

- Total Sales KPI
- Total Profit KPI
- Total Quantity KPI
- Profit Margin KPI
- Sales by Category
- Profit by Category
- Top 10 Products by Sales
- Top 10 Products by Profit
- Loss-Making Products
- Profit by Sub-Category

Conclusion

The project successfully analyzes retail business performance by identifying sales trends, profitable segments, and products causing losses. The dashboard helps businesses make data-driven decisions by highlighting areas for improvement, optimizing product strategies, and improving overall profitability.